Lecture 27

Fundamentals of Numerical Analysis

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Review of Lecture 26: Eigenvalue Methods

- **PageRank:** solve $x = Ax$, where A is the column-stochastic Google matrix; $\lambda_1 = 1$.
- **Power Method:** $x^{(k+1)} = Ax^{(k)}$ (normalized each step) converges to the dominant eigenvector; $r \to \lambda_1$ at rate $|\lambda_2/\lambda_1|^k$.
- **Inverse Power Method:** apply the power method to A^{-1} (via $A = LU$) to get the smallest eigenvalue, $r \to 1/\lambda_n$.

Today: the Singular Value Decomposition, which works for *any* $m \times n$ matrix.

SVD: motivation

SVD uses in practice:

1. Search Technology: find closely related documents or images in a database
2. Clustering: aggregate documents or images into similar groups
3. Compression: efficient image storage
4. Principal axis: find the main axis of a solid (engineering/graphics)
5. Summaries: Given a textual document, ascertain the most representative tags
6. Graphs: partition graphs into subgraphs (graphics, analysis)

SVD: Singular Value Decomposition

SVD takes an $m \times n$ matrix A and factors it:

$$A = USV^T$$

where U ($m \times m$) and V ($n \times n$) are orthogonal and S ($m \times n$) is diagonal.

Definition

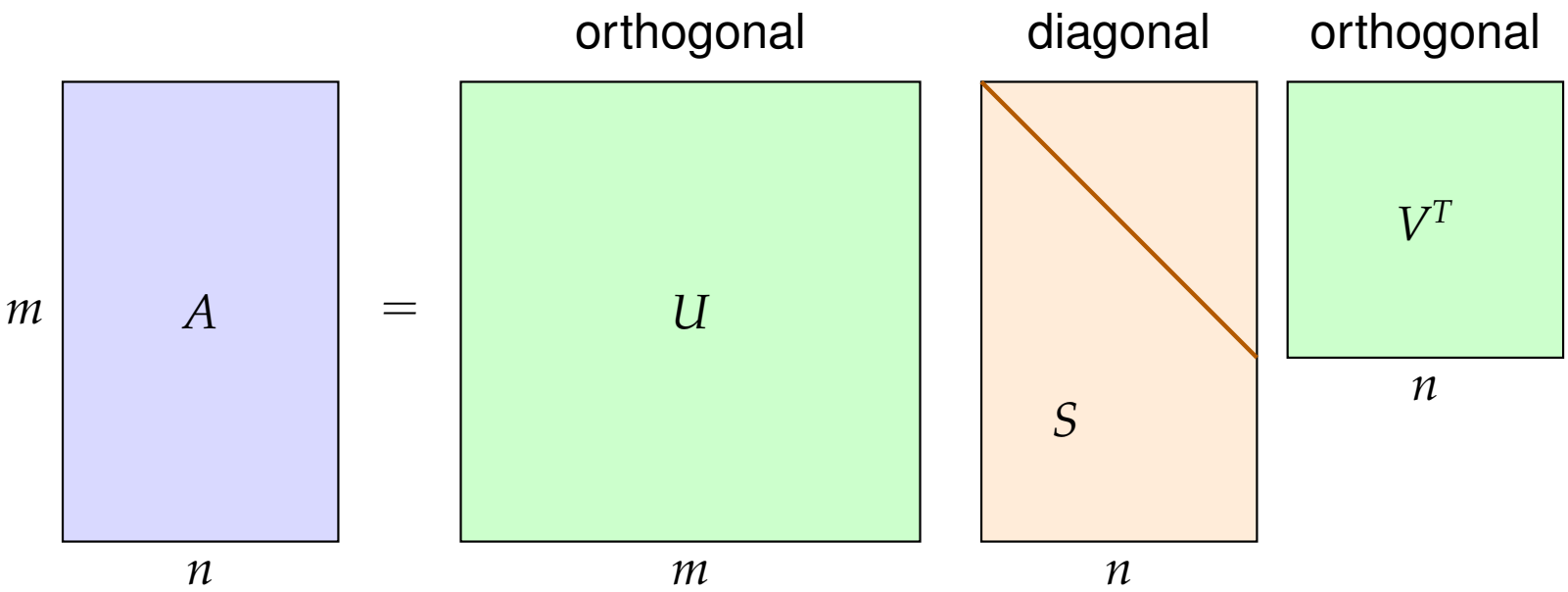
A square matrix Q is orthogonal if $Q^TQ = QQ^T = I$.

S is made up of "singular values":

$$\sigma_1 \geqslant \sigma_2 \geqslant \cdots \geqslant \sigma_r > \sigma_{r+1} = \cdots = \sigma_p = 0$$

Here, $r = \text{rank}(A)$ and $p = \min(m, n)$.

Visualization: shapes in $A = USV^T$



(drawn for $m > n$; diagonal entries $\sigma_1 \geqslant \cdots \geqslant \sigma_p \geqslant 0$, $p = \min(m, n)$)

Diagonalizing a matrix

We want to factorize A into U, S, and V^T. First step: find V. Consider

$$A = USV^T$$

and multiply by A^T from the left

$$A^TA = (USV^T)^T(USV^T) = VS^TU^TUSV^T$$

Since U is orthogonal

$$A^TA = VS^TSV^T = VS^2V^T$$

where S^2 denotes the diagonal matrix S^TS. This is called a similarity transformation.

Definition

Matrices A and B are similar if there is an invertible matrix Q such that

$$Q^{-1}AQ = B$$

Theorem

Similar matrices have the same eigenvalues.

Proof

Let $B = Q^{-1}AQ$ and suppose $Bv = \lambda v$.

$$\begin{aligned} Bv &= \lambda v \\ Q^{-1}AQv &= \lambda v \\ AQv &= \lambda Qv \\ Aw &= \lambda w, \qquad w = Qv. \end{aligned}$$

So λ is an eigenvalue of A as well. Further, if v is an eigenvector of B, Qv is an eigenvector of A.

So far...

Need $A = USV^T$

Look for V such that $A^TA = VS^2V^T$. Here S^2 is diagonal.

Since V is orthogonal, $V^T = V^{-1}$, so A^TA and S^2 are similar and have the same eigenvalues. So the diagonal matrix S^2 holds the eigenvalues of A^TA (that is, $\sigma_i^2 = \lambda_i(A^TA)$) and V is the matrix of eigenvectors.

To see the latter, write $A^TA\,V = VS^2$. Column i of this equation reads $A^TA\,v_i = \sigma_i^2 v_i$, where $v_i = Ve_i$ is the ith column of V.

Similarly...

Now consider

$$A = USV^T$$

and multiply by A^T from the right

$$AA^T = (USV^T)(USV^T)^T = USV^TVS^TU^T$$

Since V is orthogonal

$$AA^T = USS^TU^T = US^2U^T$$

Now U is the matrix of eigenvectors of AA^T (with the same nonzero eigenvalues σ_i^2).

In the end...

We get

$$A = \begin{bmatrix} \vdots & \vdots & \vdots \\ u_1 & \cdots & u_m \\ \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} \sigma_1 & & & & \\ & \ddots & & & \\ & & \sigma_r & & \\ & & & \ddots & \\ & & & & 0 \end{bmatrix} \begin{bmatrix} \cdots & v_1^T & \cdots \\ \cdots & \vdots & \cdots \\ \cdots & v_n^T & \cdots \end{bmatrix}$$

Example

Decompose

$$A = \begin{bmatrix} 2 & -2 \\ 1 & 1 \end{bmatrix}$$

First construct $A^T A$:

$$A^T A = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix}$$

Eigenvalues: $\lambda_1 = 8$ and $\lambda_2 = 2$. So

$$S^2 = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix} \quad \Rightarrow \quad S = \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}$$

Example

Now find V and U. The columns of V are the (normalized) eigenvectors of A^TA.

- $\lambda_1 = 8$: $(A^TA - \lambda_1 I)v_1 = 0$

$$\Rightarrow \begin{bmatrix} -3 & -3 \\ -3 & -3 \end{bmatrix} v_1 = 0 \Rightarrow \quad \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} v_1 = 0 \Rightarrow \quad v_1 \propto \begin{bmatrix} -1 \\ 1 \end{bmatrix} \longrightarrow v_1 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

- $\lambda_2 = 2$: $(A^TA - \lambda_2 I)v_2 = 0$

$$\Rightarrow \begin{bmatrix} 3 & -3 \\ -3 & 3 \end{bmatrix} v_2 = 0 \Rightarrow \quad \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} v_2 = 0 \Rightarrow \quad v_2 \propto \begin{bmatrix} 1 \\ 1 \end{bmatrix} \longrightarrow v_2 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

- Finally:

$$V = \begin{bmatrix} -\sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$$

Example

Now find U. The columns of U are the eigenvectors of $AA^T = \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix}$.

- $\lambda_1 = 8$: $(AA^T - \lambda_1 I)u_1 = 0$

$$\Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & -6 \end{bmatrix} u_1 = 0 \quad \Rightarrow \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} u_1 = 0 \quad \Rightarrow \quad u_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

- $\lambda_2 = 2$: $(AA^T - \lambda_2 I)u_2 = 0$

$$\Rightarrow \begin{bmatrix} 6 & 0 \\ 0 & 0 \end{bmatrix} u_2 = 0 \quad \Rightarrow \quad \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} u_2 = 0 \quad \Rightarrow \quad u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- The sign of each u_i must be consistent with v_i: use $u_i = Av_i/\sigma_i$. For instance, $Av_1 = [-2\sqrt{2},\ 0]^T$, so $u_1 = [-1,\ 0]^T$.
- Finally:

$$U = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

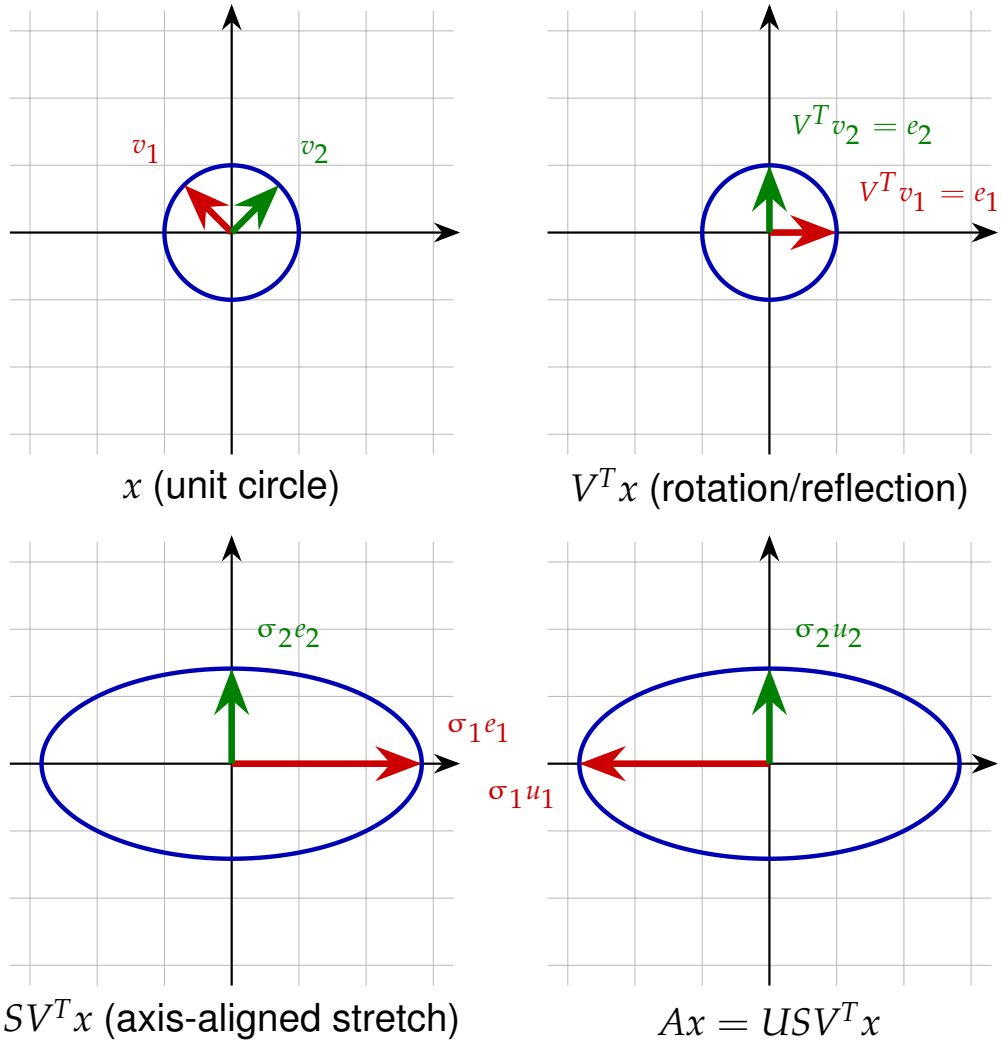
Example

Together, $A = USV^T$:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} -\sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ 1 & 1 \end{bmatrix}$$

Here the last factor is V^T; it happens to equal V because V is symmetric.

Visualization: the unit circle under $A = USV^T$



SVD: who cares?

How can we actually *use* $A = USV^T$? We can use this to represent A with far fewer entries...

Notice what $A = USV^T$ looks like:

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T + 0 u_{r+1} v_{r+1}^T + \cdots + 0 u_p v_p^T$$

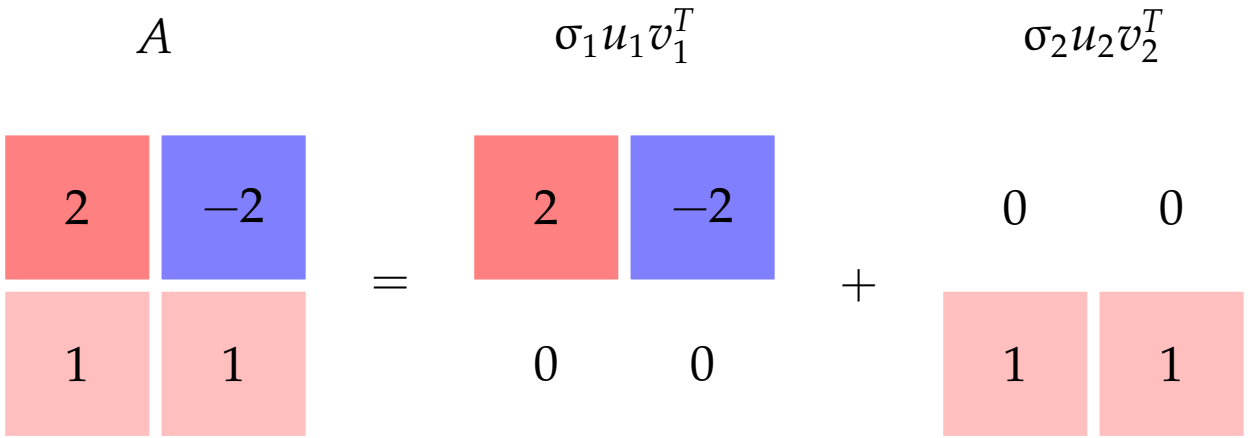
This is easily truncated to the $k \leqslant r$ largest terms:

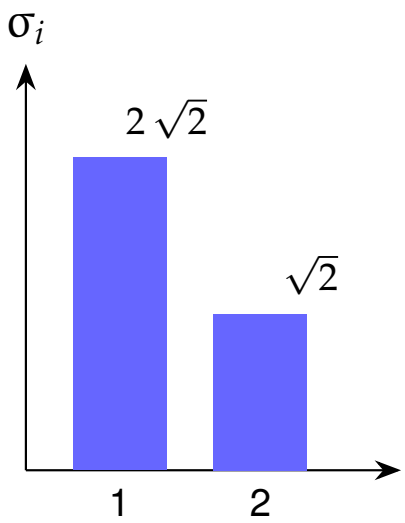
$$A \approx A_k = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_k u_k v_k^T$$

see `svd_test.m`
What are the savings?

- A takes $m \times n$ storage
- using k terms of U and V takes $k(1 + m + n)$ storage

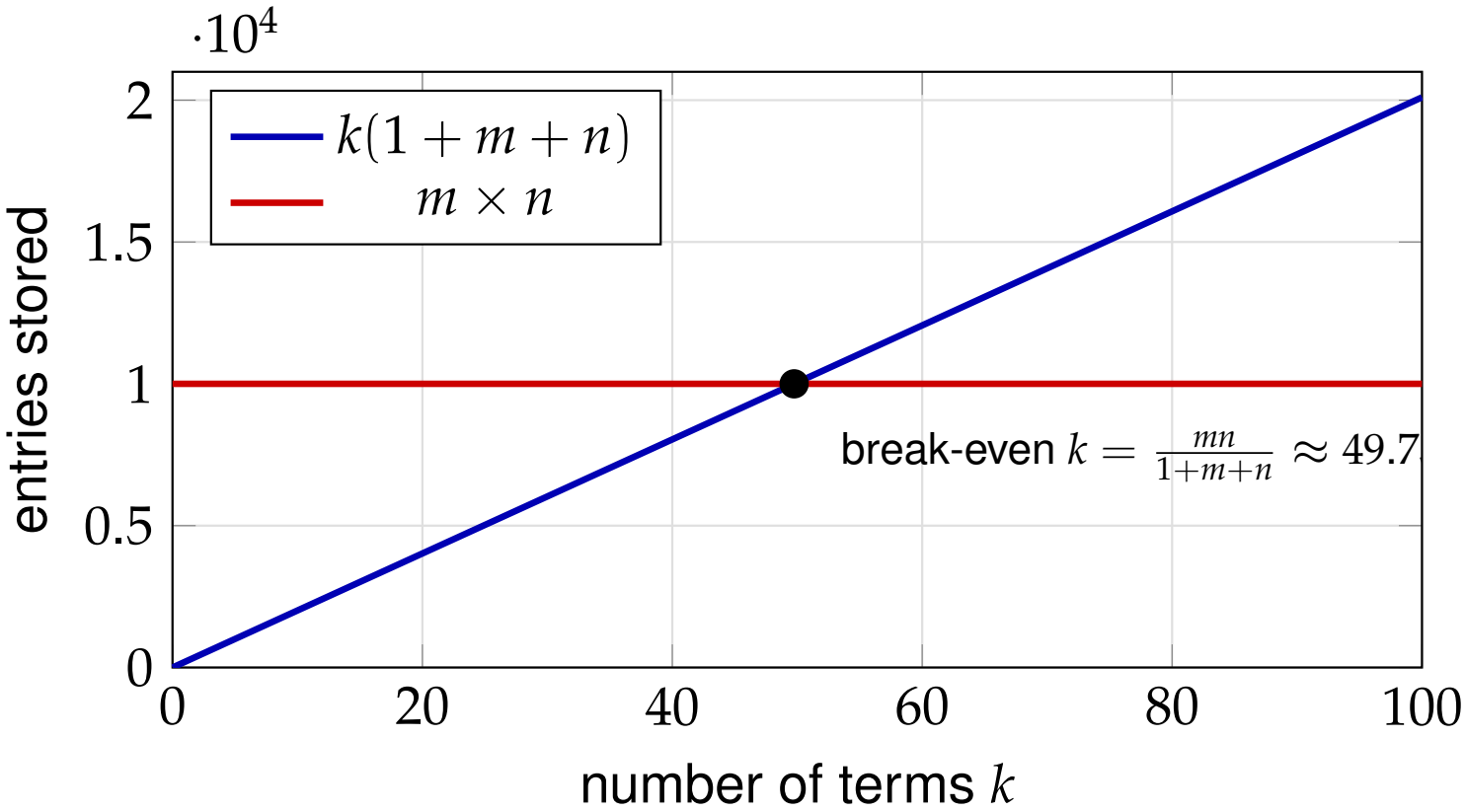
Visualization: $A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T$ for the example





Each term has rank 1; truncating at $k = 1$ leaves error $\|A - A_1\|_2 = \sigma_2 = \sqrt{2}$.

Visualization: storage of A_k versus A (here $m = n = 100$)



Motivation for Low-Rank Approximation

Many real-world matrices are *approximately low rank*:

$$A \approx A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T, \qquad k \ll \min(m, n)$$

- Data often lie near a low-dimensional subspace (e.g. correlated features).
- A_k captures the most energetic directions of A.
- Greatly reduces storage and computation while preserving structure.

Example applications:

- Image compression (store top singular triplets).
- Recommendation systems (latent factors).
- Reduced-order models for PDEs or dynamical systems.

Energy and Error in Low-Rank SVD

The total energy (Frobenius norm squared) of A is

$$\|A\|_F^2 = \sum_{i=1}^{r} \sigma_i^2.$$

Keeping k modes captures fraction

$$E_k = \frac{\sum_{i=1}^{k} \sigma_i^2}{\sum_{i=1}^{r} \sigma_i^2}.$$

- Choose k such that $E_k \geqslant 0.95$ or 0.99.
- Error in 2-norm: $\|A - A_k\|_2 = \sigma_{k+1}$.
- Error in Frobenius norm: $\|A - A_k\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2$.

Interpretation: each additional singular value adds an orthogonal "direction of variance."

Geometric Picture of Low-Rank Approximation

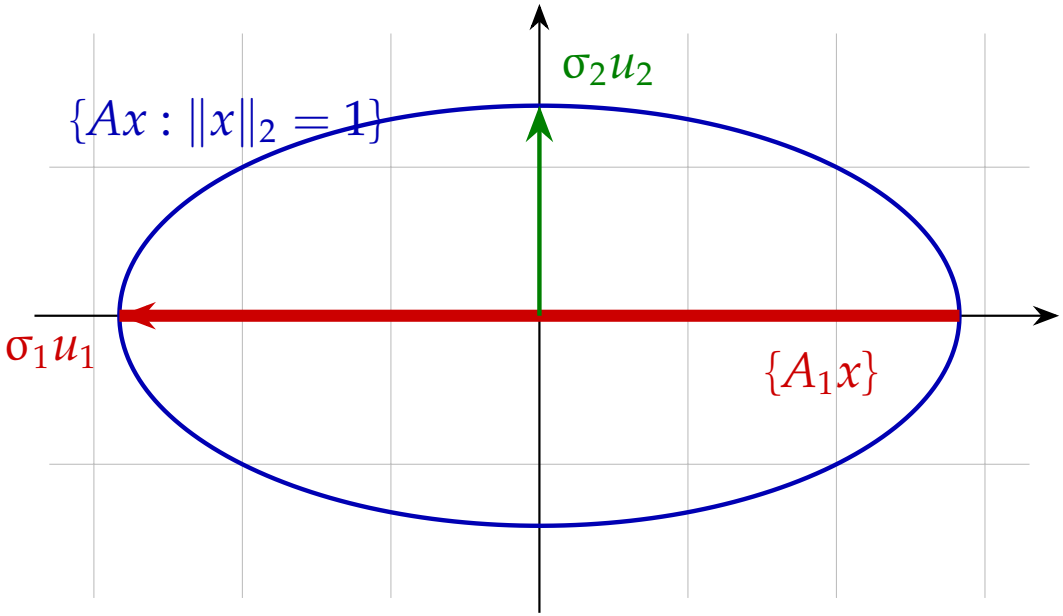
- A maps the unit sphere to an ellipsoid with axes σ_i.
- Truncating SVD removes the smallest axes $\sigma_{k+1}, \ldots$, flattening the ellipsoid into a lower-dimensional subspace.
- The approximation A_k projects data onto the span of the top k left singular vectors U_k.

$$A \approx U_k \Sigma_k V_k^T, \quad A_k x = U_k \Sigma_k (V_k^T x)$$

Meaning:

- V_k encodes *dominant input directions*.
- U_k encodes *dominant output directions*.

Visualization: truncation $A \to A_1$ for the example



$$\|A - A_1\|_2 = \sigma_2 = \sqrt{2}, \quad E_1 = \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} = \frac{8}{10}$$

Least Squares: Background

We often need to solve an overdetermined system

$$Ax \approx b, \qquad A \in \mathbb{R}^{m \times n},\ m > n.$$

Goal: minimize the residual

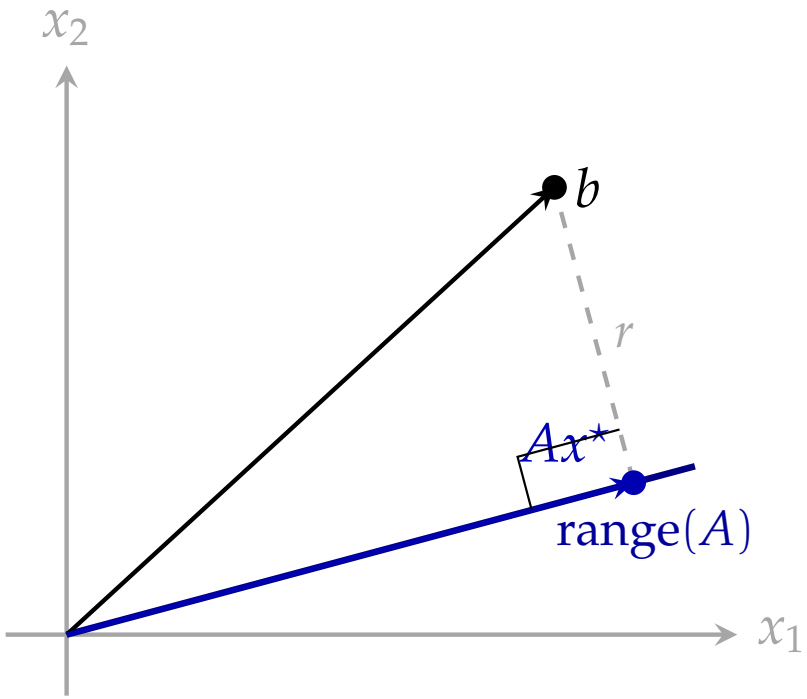
$$\min_x \|Ax - b\|_2.$$

- No exact solution in general if $b \notin \text{range}(A)$.
- The least-squares solution satisfies the **normal equations**

$$A^T A x = A^T b.$$

- $A^T A$ is symmetric positive semidefinite; invertible if $rank(A) = n$.

Geometric Interpretation of Least Squares



b is projected orthogonally onto $\text{range}(A)$*:* $b = Ax^\star + r$*, with* $r \perp \text{range}(A)$.

SVD Solution to Least Squares

Let $A = U\Sigma V^T$. Then $A^TA = V\Sigma^2 V^T$, and the normal equations give

$$x^\star = (A^TA)^{-1}A^Tb = V\Sigma^{-1}U^Tb.$$

Compact form

$$x^\star = A^+b, \quad A^+ = V\Sigma^+U^T.$$

- Σ^+ inverts only nonzero singular values.
- Works even if A is rank deficient.
- Minimum–norm solution among all least-squares minimizers.

SVD-Based Regularization

For ill-conditioned A, small σ_i amplify noise.

$$x^\star = \sum_{i=1}^{r} \frac{u_i^T b}{\sigma_i} v_i$$

- If some σ_i are tiny, $u_i^T b/\sigma_i$ becomes unstable.
- **Truncated SVD (TSVD):**

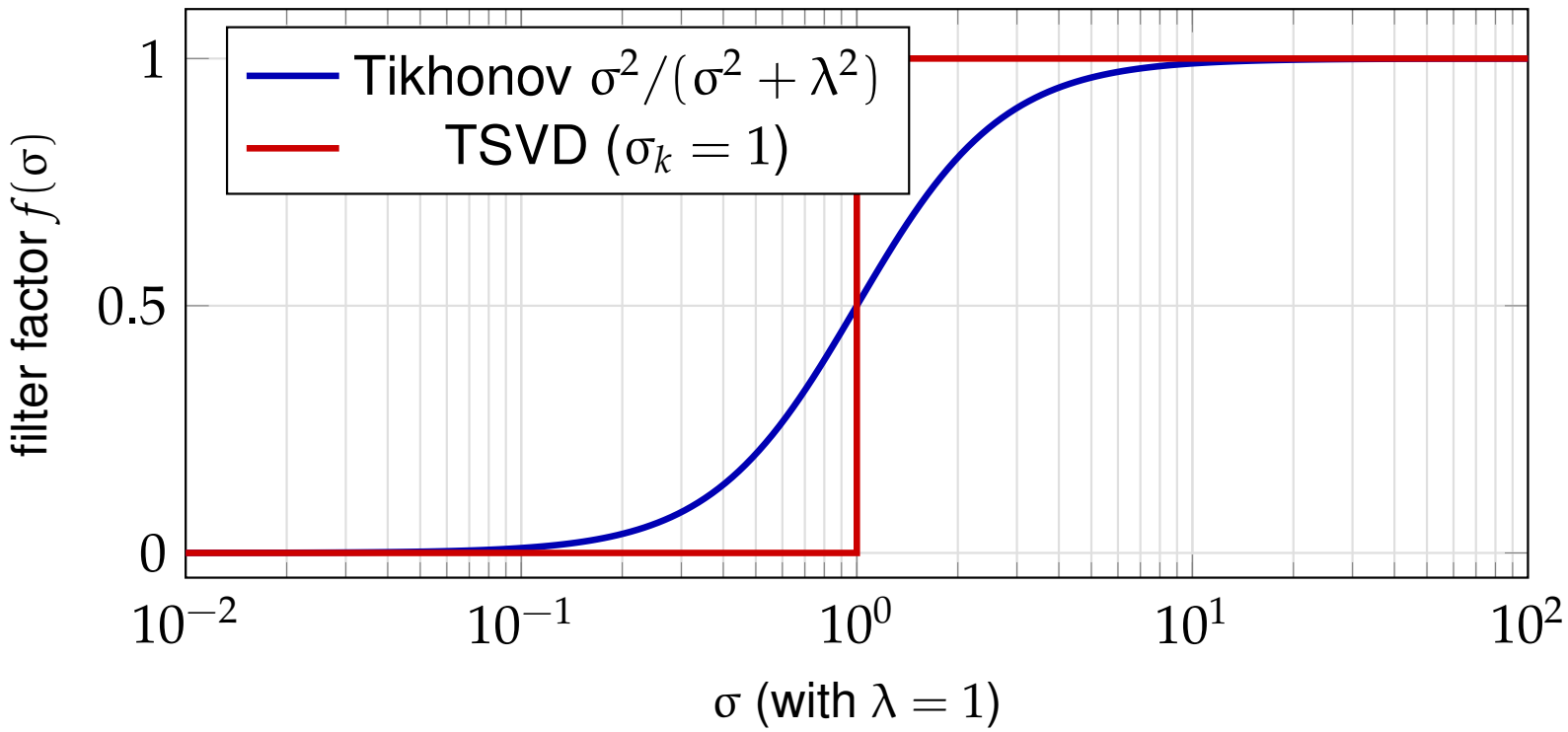
$$x_k = \sum_{i=1}^{k} \frac{u_i^T b}{\sigma_i} v_i.$$

- **Tikhonov (ridge) regularization:**

$$x_\lambda = \sum_{i=1}^{r} \frac{\sigma_i}{\sigma_i^2 + \lambda^2} (u_i^T b) v_i.$$

- Both control noise amplification via spectral filtering.

Visualization: spectral filters of TSVD and Tikhonov



Low-Rank Approximation and Least Squares Connection

- The rank-k SVD A_k provides a reduced model for A.
- Solve a smaller LS problem:

$$x_k = \arg\min_x \|A_k x - b\|_2.$$

- Equivalent to projecting data onto dominant subspace spanned by V_k:

$$x_k = V_k(\Sigma_k^{-1} U_k^T b).$$

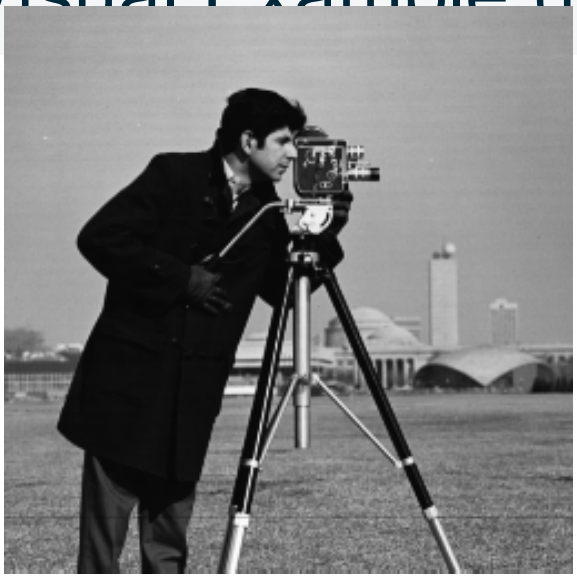
- Greatly reduces cost and improves conditioning when small σ_i cause numerical instability.

Why SVD for Images?

- A grayscale image is just a big table of numbers (pixels) $\Rightarrow$ a matrix A.
- **Goal:** keep the important visual information using fewer numbers (*compression*).
- **SVD idea (informal):** A can be written as a sum of simple "layers" (patterns).
- If we keep only the first k layers, we get a good approximation of the image.

Key benefit: smaller storage and faster processing, with little loss in quality (for reasonable k).

Visual Example (fill these figures)



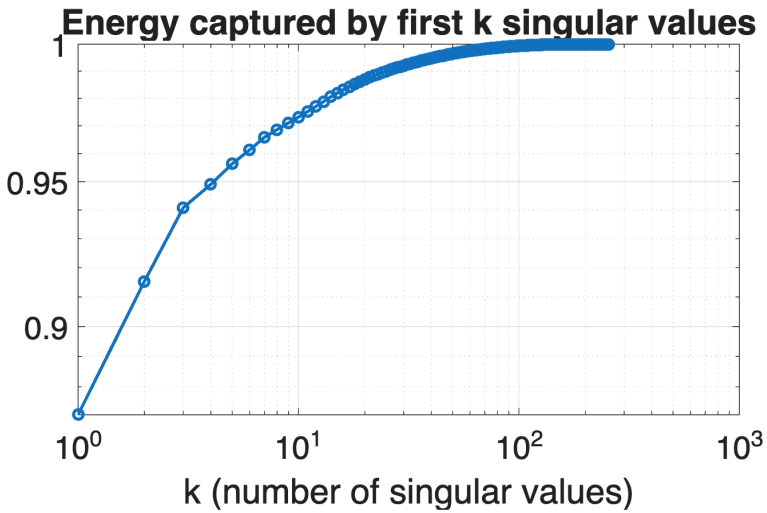
Original



Rank-10



Rank-40



Singular value



Rank-80

Minimal Code to Generate the Figures (MATLAB)

```
I = im2double(imread('cameraman.tif'));   % grayscale image
[U,S,V] = svd(I,'econ'); s = diag(S);     % SVD and singular values

% helper to save rank-k reconstruction
save_k = @(k,name) imwrite(U(:,1:k)*S(1:k,1:k)*V(:,1:k)', ...
                           fullfile('figs',name));
if ~exist('figs','dir'), mkdir figs; end
imwrite(I, 'figs/svd_img_original.png');
save_k(20,'svd_img_k20.png'); save_k(60,'svd_img_k60.png');
save_k(100,'svd_img_k100.png');

Ek = cumsum(s.^2)/sum(s.^2);                % cumulative energy
f = figure('visible','off'); plot(Ek,'-o'); grid on
xlabel('k'); ylabel('Energy E_k'); ylim([0 1])
print(f, 'figs/svd_sigma_energy.png','-dpng','-r200'); close(f)
```

Takeaways

- SVD breaks an image into simple patterns, ranked by importance.
- Keeping only the first k patterns gives a smaller file with similar quality.
- Use the energy plot to pick k (e.g., 95%–99%).
- This same idea powers *PCA*, *denoising*, and *model reduction*.